DATA VISUALIZATION
MINI PROJECT REPORT

Car Sales Interactive Dashboard

(Built using Microsoft Power BI Desktop)

Submitted by:

Omkar Jadhawar :- 202501110111

Aman Shaikh :- 202501110112

Div :- AL2 – A21

Branch :- AIML

Video link :-

https://drive.google.com/file/d/1C7WSmi7qoLbS5HcDoHIpFWzwNyXxxTvD/view?usp=drivesdk

Project link :-

https://app.powerbi.com/groups/me/reports/830720af-6298-4454-a326-9adcac0826a2/1222ab60cd891cb44061?experience=power-bi

1. Introduction

In this project we made a Car Sales Dashboard using Microsoft Power BI Desktop. The main idea was to take the raw sales data and show it in a way that is easy to understand by using different charts and graphs. This dashboard has 3 pages and each page shows different kind of information about the car sales.

The dataset we used had around 23,906 records of car sales from year 2022 and 2023. It has details like customer name, car company, dealer name, region, price and many more things. We connected all three pages using navigation buttons and slicers so that user can easily move from one page to another.

Project Objective:

The main objective of this project is to build a 3 page interactive dashboard in Power BI which can help business people like sales managers or analysts to see how the car sales are going. It gives information about revenue, dealer performance, region wise sales and also customer details.

2. Dataset Overview

The dataset used in this project is Car_Sales.xlsx. It contains transactional data for a car dealership. We loaded this data into Power BI using the Excel connector. Before making any visuals we first cleaned the data in Power Query Editor.

2.1 Data Structure

Below table shows the columns present in the dataset and what each column means –

Column Name	Data Type	Description
Car_id	Text	Unique id for each sale transaction
Date	Date	Date when sale happened
Customer Name	Text	Name of the customer who bought the car
Gender	Text	Male or Female
Annual Income	Number	Income of customer in USD
Dealer_Name	Text	Name of the dealer
Company	Text	Car brand like BMW, Ford etc.
Model	Text	Which model of the car was sold
Engine	Text	Type of engine
Transmission	Text	Auto or Manual
Color	Text	Color of the car
Price ($)	Number	Sale price in USD
Dealer_No	Text	Dealer's ID number
Dealer_Region	Text	Region where dealer is located
Body Style	Text	SUV, Sedan, Hatchback etc.
Phone	Text	Customer phone number (not used in analysis)

2.2 Data Preparation in Power Query

We did the following data cleaning steps before building the dashboard -

- Date column was in Text format so we changed it to Date type so that we can use it for month and year wise grouping.
- Price ($) and Annual Income columns were changed to Decimal Number so that we can do SUM and AVERAGE on them.
- Engine column had some wrong text like 'DoubleA Overhead Camshaft' so we used Replace Values to fix it.
- Phone column was removed because it was not useful for our analysis.
- We also created a separate Calendar Table using the formula CALENDAR(MIN(car_data[Date]), MAX(car_data[Date])) and added Month, Week and Year columns to it. This table is connected to car_data table through the Date column.

3. Data Model & Relationships

We used a Star Schema model in this project. The car_data table is our main fact table and the Calendar Table is the dimension table. Below is the relationship between them –

Fact Table	Dimension Table	Relationship	Cardinality
car_data	Calendar Table	car_data[Date] → Calendar[Date]	Many-to-One (*:1)

The filter direction is set to Single which means filters go from Calendar Table to car_data. This is important because all the time based slicers and visuals need to work correctly.

4. DAX Measures

We created the following DAX measures to calculate the KPIs and other values used in our dashboard –

Measure Name	DAX Formula
Total Revenue	SUM(car_data[Price ($)])
Total Cars Sold	COUNTROWS(car_data)
Avg Car Price	AVERAGE(car_data[Price ($)])
Avg Customer Income	AVERAGE(car_data[Annual Income])
Total Dealers	DISTINCTCOUNT(car_data[Dealer_Name])

Top Region Revenue	MAXX(VALUES(car_data[Dealer_Region]), CALCULATE([Total Revenue]))
Revenue Per Dealer	DIVIDE([Total Revenue], [Total Dealers], 0)
Total Customers	DISTINCTCOUNT(car_data[Customer Name])
Male Sales %	DIVIDE(CALCULATE([Total Cars Sold], car_data[Gender]="Male"), [Total Cars Sold], 0)
Female Sales %	DIVIDE(CALCULATE([Total Cars Sold], car_data[Gender]="Female"), [Total Cars Sold], 0)

5. Dashboard Pages

Our dashboard is divided into 3 pages. All pages are connected through navigation buttons, slicers and drill-through so user can easily go from one page to another without any confusion.

Page 1 – Sales Overview Dashboard

5.1 Purpose & Story

This is the first page of our dashboard and it shows the overall performance of the car sales business. On the top of the page we put 4 KPI cards showing Total Revenue, Total Cars Sold, Average Car Price and Average Customer Income. These are the most important numbers for any business.

We also added a line chart showing monthly revenue trend. From this chart we observed that sales are very high from October to December and very low in January and February. This makes sense because at end of year companies give offers and new models come out.

There is also a bar chart for Total Cars Sold by Company. Chevrolet, Dodge, Ford and Volkswagen have highest sales. We also added a scatter chart showing relationship between Average Income and Average Car Price which shows that richer customers buy more expensive cars. The horizontal bar chart on the right shows Revenue by Dealer Region – Austin has highest revenue of $117M.

5.2 Dashboard Screenshot – Page 1

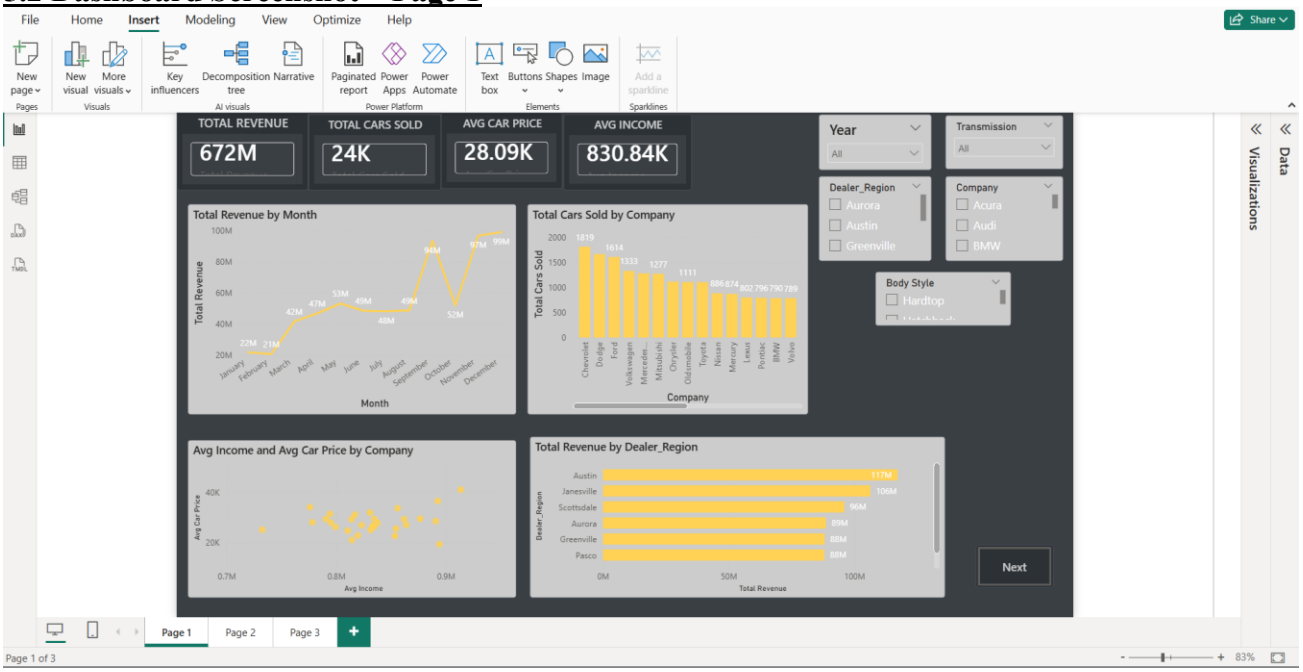

Figure 1: Page 1 – Sales Overview Dashboard showing KPI cards, Monthly Revenue Trend, Sales by Company, Income vs Price Scatter and Revenue by Dealer Region

5.3 Visuals & Configuration

Visual Type	X / Axis	Y / Value	Purpose
KPI Cards (4 nos.)	—	Total Revenue, Cars Sold, Avg Price, Avg Income	Shows main KPIs at top
Line Chart	Month	Total Revenue	Monthly trend of revenue
Bar Chart	Company	Total Cars Sold	Which brand sold most
Scatter Chart	Avg Income	Avg Car Price by Company	Income vs price relation
Horizontal Bar Chart	Total Revenue	Dealer Region	Region wise revenue
Slicers (4 nos.)	Year, Dealer Region, Transmission, Body Style, Company	—	For filtering data

5.4 Key Findings from Page 1

- Total revenue is 672M from 24K car sales.
- Average car price is $28.09K which shows mid range cars are selling most.

- Revenue is highest in October to December months.
- Chevrolet has most sales followed by Dodge and Ford.
- Austin region gives highest revenue of around $117M.
- High income customers buy high priced cars – this is seen in scatter chart.

Page 2 – Dealer & Region Analysis Dashboard

5.5 Purpose & Story

Page 2 is for deeper analysis of dealers and regions. It answers questions like which dealers are performing best and how are different body styles doing in different regions. This page is mainly useful for regional managers and operations team.

KPI cards on top show 28 total dealers, top region revenue of $117M, average revenue per dealer is $23.98M and average car price of $28.09K. Then there is a bar chart showing Revenue by Dealer Name – Rabun Used Car Sales, Progress and U-Haul CO are the top performers.

The donut chart shows total cars sold by each dealer region. All regions have almost equal share which means no single region is dominating. The stacked bar chart is very useful – it shows body style distribution by region. SUVs are most popular in Austin and Janesville regions.

5.6 Dashboard Screenshot – Page 2

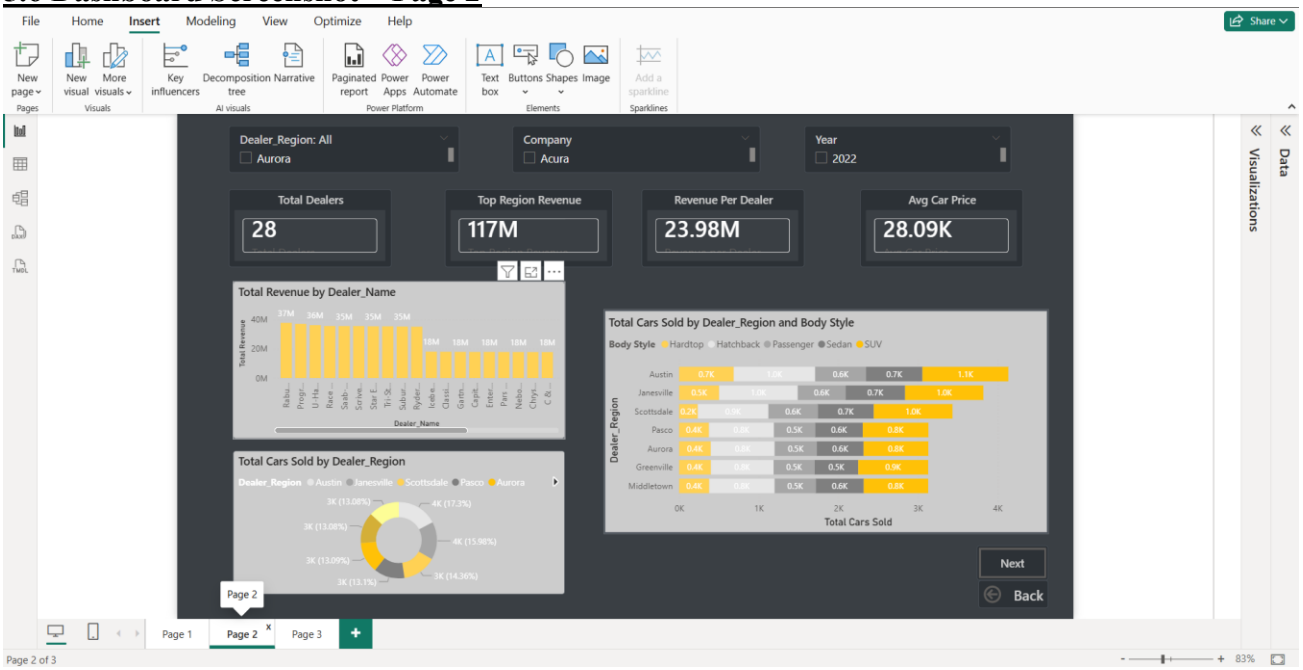

Figure 2: Page 2 – Dealer & Region Analysis showing Total Dealers KPIs, Revenue by Dealer Name, Sales by Region Donut and Stacked Body Style breakdown

5.7 Visuals & Configuration

Visual Type	X / Axis	Y / Value	Purpose
KPI Cards (4 nos.)	—	Total Dealers, Top Region Revenue, Rev/Dealer, Avg Price	Dealer network summary
Bar Chart	Dealer Name	Total Revenue (Top N)	Which dealer earns most
Donut Chart	Dealer Region	Total Cars Sold	Region wise market share
Stacked Bar Chart	Total Cars Sold	Dealer Region (Body Style)	Body style mix by region
Slicers (3 nos.)	Dealer Region, Company, Year	—	For filtering

5.8 Key Findings from Page 2

- There are 28 dealers across 7 different regions.
- Top dealers earn much more than average dealers.
- All regions have more or less equal share (around 13-17% each).
- SUVs are sold most in Austin and Janesville regions.
- Hatchbacks and Sedans are distributed equally across all regions.
- Revenue per dealer is $23.98M which can be used as a benchmark.

Page 3 – Customer Insights Dashboard

5.9 Purpose & Story

Page 3 is all about customers – who is buying and what they like. This page is for marketing team and CRM analysts. We have 3 slicers on this page for Gender, Dealer Region and Income Range so that user can filter customers easily.

The KPI cards show Average Income is $830.84K, Average Car Price is $28.09K, Male customers are 79% and Female are 21%. There is also a scatter chart comparing income vs car price by company and gender. Interestingly female customers buy cars at similar price range as male customers even though income is slightly lower.

Pie chart confirms 78.63% cars are bought by male customers. Bar chart shows Pale White is most popular car color with 11.3K units sold, followed by Black (7.9K) and Red (4.8K). At the bottom there is a column chart showing top customers by revenue which can be useful for giving special offers to VIP customers.

5.10 Dashboard Screenshot – Page 3

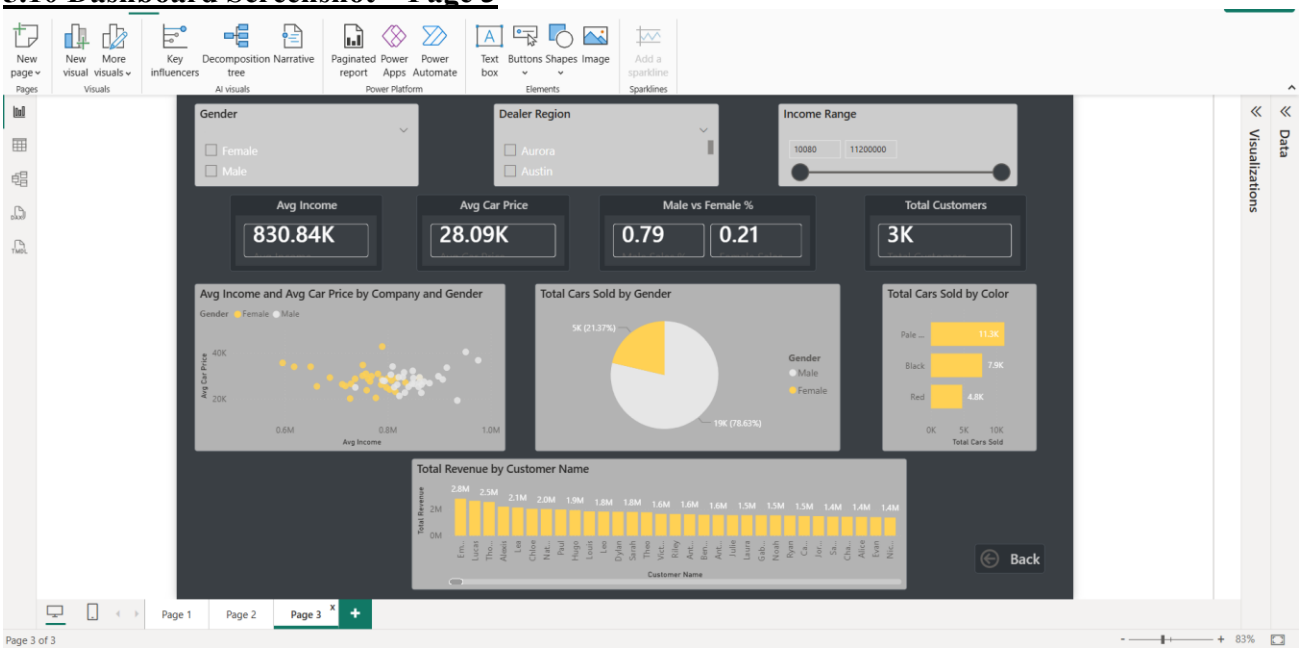

Figure 3: Page 3 – Customer Insights Dashboard showing Income vs Price Scatter, Gender Pie Chart, Sales by Color and Top Customers by Revenue

5.11 Visuals & Configuration

Visual Type	X / Axis	Y / Value	Purpose
KPI Cards (4 nos.)	—	Avg Income, Avg Price, Male %, Female %, Total Customers	Customer overview
Scatter Chart	Avg Income	Avg Car Price by Company & Gender	Income vs price by gender
Pie Chart	Gender	Total Cars Sold	Male vs Female split
Horizontal Bar Chart	Total Cars Sold	Car Color	Most popular colors
Column Chart	Customer Name	Total Revenue	Top customers by revenue
Slicers (3 nos.)	Gender, Dealer Region, Income Range	—	For customer segmentation

5.12 Key Findings from Page 3

- 78.63% of car buyers are male customers.
- Average customer income is $830.84K which means mostly rich people are buying cars.

- Female customers pay similar price for cars as male customers.
- Pale White is the most preferred color with 11.3K units.
- Top individual customers spend up to $2.8M – they should be given VIP treatment.
- Income Range slicer helps to see difference between high and premium income buyers.

6. Dashboard Navigation & Interlinking

All 3 pages of our dashboard are linked together. We used navigation buttons, synchronized slicers and drill-through to connect them properly.

6.1 Navigation Buttons

We added Next and Back buttons on each page in the bottom right corner. These buttons use Power BI's Page Navigation feature –

- Page 1 has Next button which goes to Page 2.
- Page 2 has both Next button (goes to Page 3) and Back button (goes to Page 1).
- Page 3 has only Back button which returns to Page 2.

6.2 Synchronized Slicers

Year and Dealer_Region slicers are synchronized on all 3 pages using View > Sync Slicers in Power BI. So if you select a year on Page 1 it automatically applies to Page 2 and Page 3 also. This is very helpful for consistent analysis.

6.3 Drill-Through

We configured drill-through on Page 2 using the Dealer_Name field. So if on Page 1 you right click on any dealer in the bar chart, you can directly go to Page 2 filtered for that dealer. Power BI automatically adds a Back button on that page.

7. Design & Formatting

We tried to keep a consistent look on all 3 pages. The design choices we made are –

- Background color is dark navy (#1F3864) with gold/amber (#C9A84C) colored charts to give a professional automotive look.
- All headings are in bold white text and data labels are kept at 11-12pt size for readability.
- KPI Cards are all same size with big bold numbers so they can be read at a glance.
- Slicers are placed at top of each page in dropdown style so user always knows where to filter.
- All charts have rounded borders (4px) to look modern.

- Page titles are clearly written at the top of each page.

8. Conclusion

In this mini project we successfully built a 3 page interactive Car Sales Dashboard in Power BI. We started from raw Excel data, cleaned it in Power Query, created DAX measures and then built all the visuals. It was a good learning experience for us.

Page 1 gives overall sales summary, Page 2 shows dealer and region analysis, and Page 3 gives customer information. All pages are properly connected with each other. Different types of users like managers, operations team or marketing team can use different pages according to their need.

This project helped us understand how data visualization works in real world. We also learnt how to use DAX formulas, create relationships between tables and design a clean dashboard. Overall we are happy with the result.

Project Summary:

We built a fully working 3 page Car Sales Dashboard in Power BI using 23,906 rows of data. We made 10 DAX measures, created a Calendar Table, used synchronized slicers, drill-through navigation and tried our best to keep the design consistent and clean.